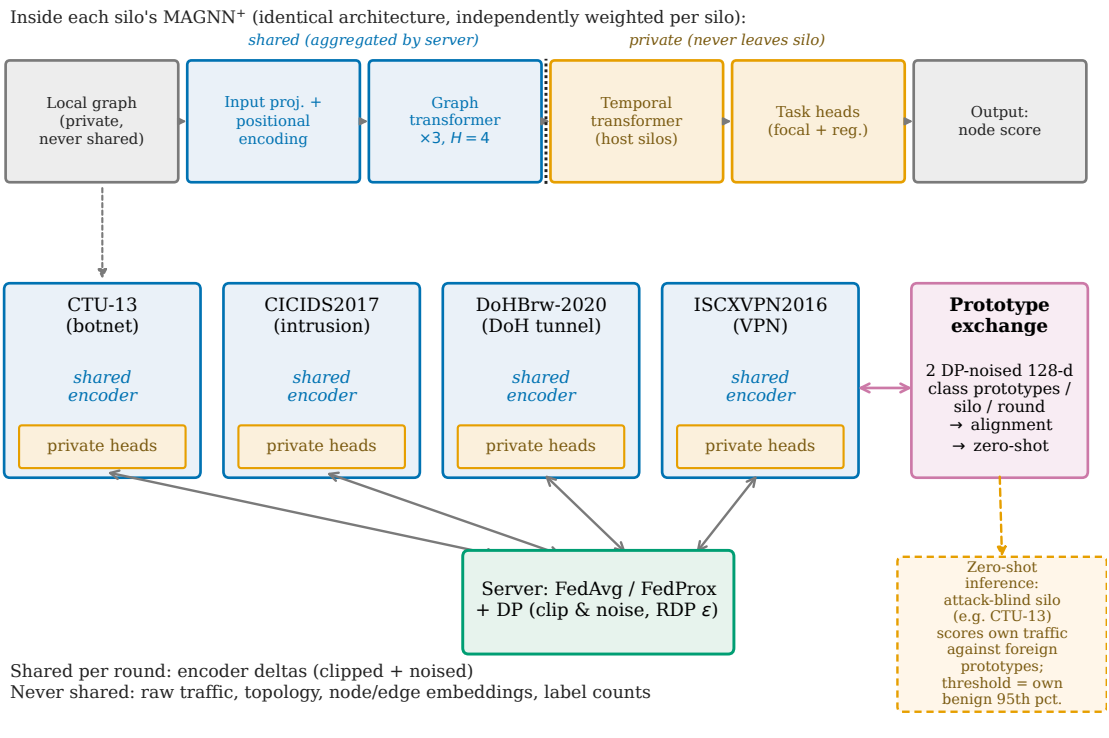
Inside each silo's MAGNN+ (identical architecture, independently weighted per silo):
shared (aggregated by server)
private (never leaves silo)
Local graph (private, never shared)
Input proj. + positional encoding
Graph transformer ×3, H = 4
Temporal transformer (host silos)
Task heads (focal + reg.)
Output: node score
CTU-13 (botnet)
CICIDS2017 (intrusion)
DoHBrw-2020 (DoH tunnel)
ISCXVPN2016 (VPN)
shared encoder
private heads
Prototype exchange
2 DP-noised 128-d class prototypes / silo / round
→ alignment
→ zero-shot
Server: FedAvg / FedProx + DP (clip & noise, RDP ε)
Zero-shot inference: attack-blind silo (e.g. CTU-13) scores own traffic against foreign prototypes; threshold = own benign 95th pct.
Shared per round: encoder deltas (clipped + noised)
Never shared: raw traffic, topology, node/edge embeddings, label counts